Exploratory Data Analysis Report

E-Commerce Sales Dataset | DecodeLabs Internship – Project 2

This report presents a complete EDA of 1,200 e-commerce orders spanning January 2023 to June 2025. The analysis covers descriptive statistics, distribution patterns, outlier detection, trend analysis, and correlation findings to extract actionable business insights.

1. Dataset Overview

Metric	Value
Total Records	1,200
Total Columns	14
Date Range	Jan 2023 – Jun 2025
Missing Values	None (0 across all columns)
Numeric Columns	4 (Quantity, UnitPrice, ItemsInCart, TotalPrice)
Categorical Columns	10 (Product, Status, Payment, etc.)

The dataset is clean with no missing values, making it immediately ready for analysis.

2. Descriptive Statistics

The five-number summary below reveals the distribution shape of each numeric variable. TotalPrice has a wide spread (std = $819.86), indicating high variability in order values.

Statistic	Quantity	UnitPrice ($)	ItemsInCart	TotalPrice ($)
Count	1,200	1,200	1,200	1,200
Mean	2.95	356.41	5.49	1,053.97
Median	3.00	364.21	5.00	823.62
Std Dev	1.41	197.18	2.28	819.86
Min	1	11.39	1	11.39
Q1 (25%)	2	186.06	4	410.52
Q3 (75%)	4	521.57	7	1,578.48
Max	5	699.93	10	3,456.40

Key Observation: The mean TotalPrice ($1,053.97) is significantly higher than the median ($823.62), indicating right skew — a small number of very high-value orders pull the average upward. This means the median is a more reliable 'typical order value' for business decisions.

3. Variable Distributions

3.1 Product Distribution

Orders are distributed fairly evenly across 7 product categories, with Printer leading slightly.

Product	Orders	Total Revenue ($)	Avg Order ($)
Printer	181	195,612.61	1,080.73
Tablet	179	186,568.95	1,042.28
Chair	178	195,620.11	1,098.99
Laptop	173	192,126.56	1,110.56
Desk	170	167,459.93	985.06
Monitor	163	175,651.41	1,077.62
Phone	156	151,722.39	972.58

Insight: Laptop has the highest average order value ($1,110.56), while Phone generates the least revenue overall ($151,722) due to fewer orders. The product mix is well-balanced — no single product dominates, reducing concentration risk.

3.2 Order Status Distribution

A major red flag: 41.4% of orders are either Cancelled or Returned, indicating serious fulfillment issues.

Status	Count	Percentage
Cancelled	250	20.8%
Returned	247	20.6%
Pending	237	19.8%
Shipped	235	19.6%
Delivered	231	19.3%

Insight: Only 19.3% of orders are confirmed Delivered. Combined Cancelled + Returned = 41.4%. This is a critical business problem — nearly half of all orders do not result in successful fulfillment, representing significant lost revenue and operational cost.

3.3 Payment Method Distribution

Payment Method	Orders	Share
Online	258	21.5%
Cash	246	20.5%
Credit Card	234	19.5%
Debit Card	232	19.3%
Gift Card	230	19.2%

Insight: Payment methods are evenly spread — no single method dominates. Online payments lead marginally (21.5%). This suggests the platform serves a diverse customer base and should maintain all payment options.

3.4 Referral Source Distribution

Source	Orders	Total Revenue ($)	Avg Revenue ($)
Instagram	259	275,285.45	1,062.88
Email	250	261,808.55	1,047.23
Google	241	250,441.48	1,039.18
Facebook	228	250,410.90	1,098.29
Referral	222	226,815.58	1,021.69

Insight: Instagram drives the most orders (259) and highest total revenue ($275K). However, Facebook customers have the highest average order value ($1,098.29), suggesting they buy more expensive items. This signals that Facebook may be better for high-value campaigns.

4. Outlier Detection

Using the IQR method (flag values beyond Q1 – 1.5×IQR or Q3 + 1.5×IQR):

Parameter	Value
Q1 (25th percentile)	$410.52
Q3 (75th percentile)	$1,578.48
IQR (Q3 – Q1)	$1,167.96
Lower Fence (Q1 – 1.5×IQR)	–$1,341.41 (no lower outliers)
Upper Fence (Q3 + 1.5×IQR)	$3,330.41
Outliers Detected	8 orders above upper fence

Only 8 orders exceed the upper fence of $3,330.41. All 8 outlier orders have Quantity = 5 (the maximum), which fully explains their high value — these are legitimate bulk orders, not data errors. The range of outlier values is $3,334 to $3,456. These are SIGNALS (bulk buyers), not NOISE (data errors) — they should be investigated for a bulk/wholesale pricing strategy.

5. Trend Analysis

5.1 Revenue by Year

Year	Orders	Total Revenue ($)	Avg Order ($)
2023	510	552,643.24	1,083.61
2024	459	480,235.87	1,046.27
2025 (Jan–Jun)	231	231,882.85	1,003.82

Insight: Both order volume and average order value are declining year over year. 2024 saw a 10% drop in orders vs 2023. The declining average order value ($1,083 → $1,046 → $1,003) suggests either product mix shift toward cheaper items or increasing price sensitivity. This is a concerning trend requiring management attention.

5.2 Coupon Code Usage

Coupon Code	Orders	Share
FREESHIP	622	51.8%
WINTER15	292	24.3%
SAVE10	286	23.8%

Insight: FREESHIP is used in 51.8% of all orders — customers are highly motivated by free shipping. This indicates shipping costs may be a key barrier to purchase conversion. Consider building free shipping into pricing strategy.

6. Correlation Analysis

Variable	Quantity	UnitPrice	ItemsInCart	TotalPrice
Quantity	1.000	0.015	0.650	0.615
UnitPrice	0.015	1.000	0.001	0.717
ItemsInCart	0.650	0.001	1.000	0.393
TotalPrice	0.615	0.717	0.393	1.000

Key findings from the Pearson correlation matrix:

• UnitPrice vs TotalPrice (r = 0.717): Strong positive correlation — the price per item is the biggest driver of total order value. Higher-priced products generate much higher revenue.

• Quantity vs TotalPrice (r = 0.615): Moderate-strong correlation — ordering more units also significantly increases total spend, as expected.

• ItemsInCart vs Quantity (r = 0.650): Customers with more items in their cart tend to order higher quantities — these are likely bulk or repeat buyers.

• UnitPrice vs Quantity (r = 0.015): Near-zero — customers buy similar quantities regardless of product price. Price does NOT discourage quantity purchases.

7. Summary of Key Findings

#	Finding	Business Implication
1	Dataset is clean — 0 missing values across 1,200 rows	Data is reliable; no cleaning needed before modeling
2	TotalPrice is right-skewed (mean $1,054 > median $824)	Use median for "typical order" reporting, not mean
3	41.4% of orders are Cancelled or Returned	Critical: investigate fulfillment & return reasons urgently
4	Only 8 outliers — all legitimate bulk orders (Qty=5)	Target bulk buyers with wholesale or loyalty offers
5	Instagram drives most revenue ($275K, 259 orders)	Prioritize Instagram in marketing spend
6	Facebook has highest avg order value ($1,098)	Use Facebook for premium product campaigns
7	UnitPrice is the strongest revenue driver (r=0.717)	Push premium products to boost revenue per order
8	51.8% of orders use FREESHIP coupon	Free shipping is key to conversion — bake into pricing
9	Revenue declining YoY: $553K → $480K → $232K (half-year)	Investigate causes of order volume decline in 2024–25
10	Laptop has highest avg order value ($1,110.56)	Feature Laptop prominently to maximize revenue per sale

Report prepared as part of DecodeLabs Data Analytics Internship – Project 2 (EDA)

Dataset: E-Commerce Orders (1,200 records) | Analysis Date: June 2025